

Grade 6
Unit 11
Lesson 3 Practice

Name _____
Date _____
Period _____

1. Evaluate the expression $8b + 10 - (9 - 6)^2$ for $b = -5$.

$$\begin{aligned} &= \underbrace{8(-5)} + 10 - \underbrace{(9 - 6)}^2 \\ &= -40 + 10 - \underbrace{(3)}^2 \\ &= -40 + 10 - 9 \\ &= -30 - 9 = -39 \end{aligned}$$

2. Evaluate $-7m - 5q^2 + \frac{s}{2}$ when $m = -8$, $q = -2$, and $s = 56$.

$$\begin{aligned} &= \underbrace{-7(-8)} - 5\underbrace{(-2)^2} + \frac{56}{2} \\ &= 56 - 5(4) + 28 \\ &= 56 - 20 + 28 \\ &= 36 + 28 = 64 \end{aligned}$$

3. Evaluate $2s - 6q - f^2$ for $s = -8$, $q = 10$, and $f = -12$.

$$\begin{aligned} &= \underbrace{2(-8)} - \underbrace{6(10)} - \underbrace{(-12)^2} \\ &= -16 - 60 - (144) \\ &= -76 - 144 \\ &= -220 \end{aligned}$$

4. Evaluate $-m^3 - 3d + (n - y)$ if $m = 4$, $d = -6$, $n = -9$, and $y = 0$.

$$\begin{aligned} &= \underbrace{-(4)^3} - \underbrace{3(-6)} + \underbrace{(-9 - 0)} \\ &= -\underbrace{(64)} + 18 + (-9) \\ &= -46 - 9 \\ &= -55 \end{aligned}$$

5. Evaluate $4g^4 - \frac{z}{4}$ when $g = -4$ and $z = 24$.

$$\begin{aligned} &= 4(-4)^4 - \frac{(24)}{4} \\ &= 4(256) - \frac{24}{4} \\ &= 1024 - 6 \\ &= 1018 \end{aligned}$$

6. What is the value of $5(8x + 4) - 3m^2$ if $x = 3$ and $m = 15$?

$$\begin{aligned} &= 5(8(3) + 4) - 3(15)^2 \\ &= 5(24 + 4) - 3(225) \\ &= 5(28) - 3(225) \\ &= 140 - 675 \\ &= -535 \end{aligned}$$

7. Find the value of $-9(h - 6) + 2a - (9 - a)^5$ if $h = 6$, and $a = 12$.

$$\begin{aligned} &= -9(6 - 6) + 2(12) - (9 - 12)^5 \\ &= \cancel{-9(0)} + 24 - (-3)^5 \\ &= 24 - (-243) \\ &= 267 \end{aligned}$$

8. Evaluate the expression $p^3 - 5k(y - k)$ for $p = 6$, $k = -4$ and $y = 1$.

$$\begin{aligned} &= 6^3 - \underbrace{5(-4)}_{(-20)} \underbrace{(1 - (-4))}_{(5)} \\ &= 216 - (-20)(5) \\ &= 216 - (-100) \\ &= 316 \end{aligned}$$

9. Evaluate the expression $8 \div j - 12 \div (-10 + 4)$ if $j = -4$.

$$\begin{aligned} &= \underbrace{8 \div (-4)}_{-2} - \underbrace{12 \div (-10 + 4)}_{(-6)} \\ &= -2 - 12 \div (-6) \\ &= -2 - (-2) \\ &= 0 \end{aligned}$$

10. What is the value of the expression $4m - 10 + (a - 11) \cdot 4^3$ if $m = 5$ and $a = -1$?

$$\begin{aligned} &= 4(5) - 10 + \underbrace{(-1 - 11)}_{(-12)} \cdot 4^3 \\ &= 20 - 10 + (-12) \cdot 64 \\ &= \underbrace{20 - 10}_{10} + (-768) \\ &= 10 + (-768) \\ &= -758 \end{aligned}$$